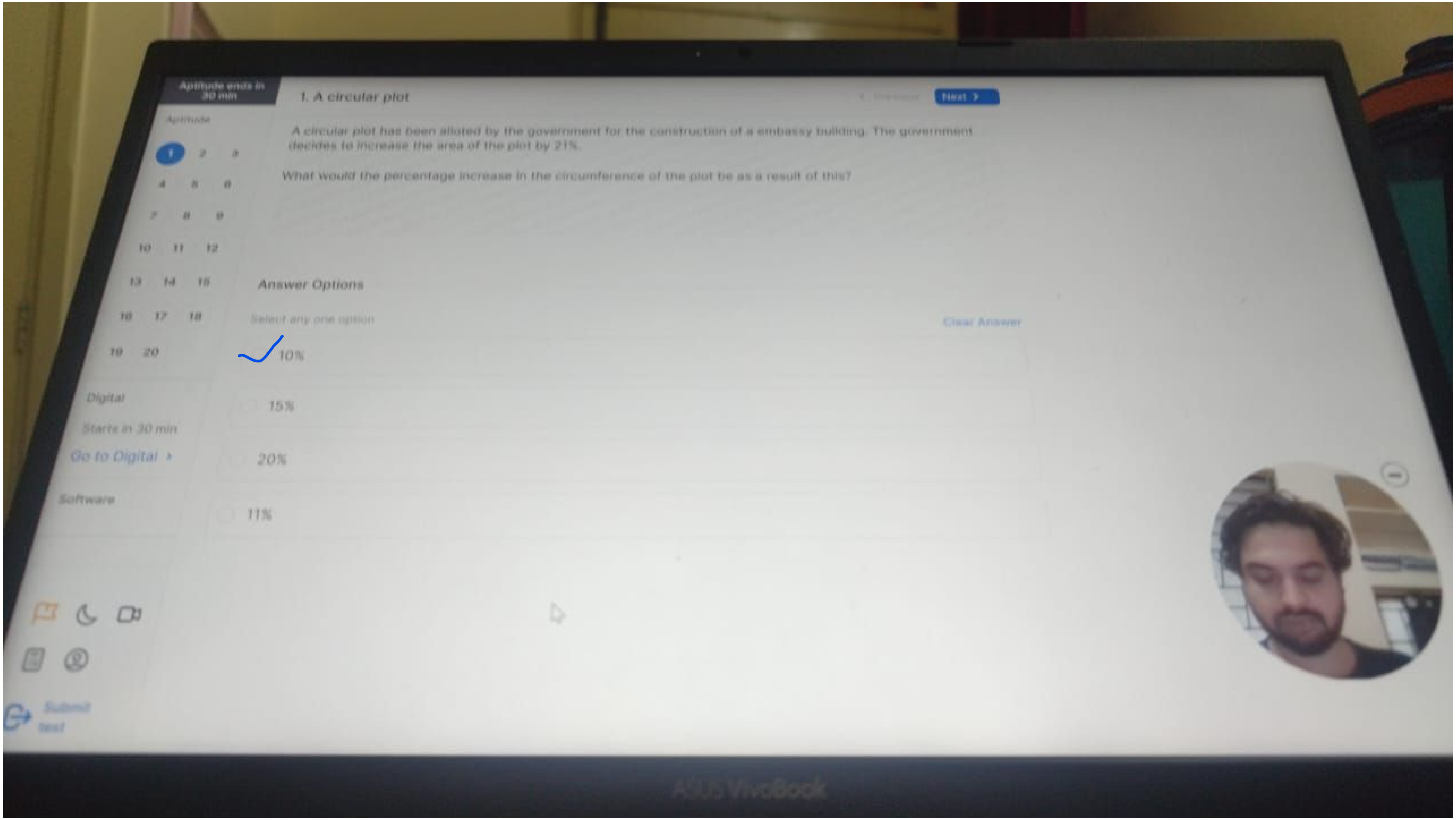
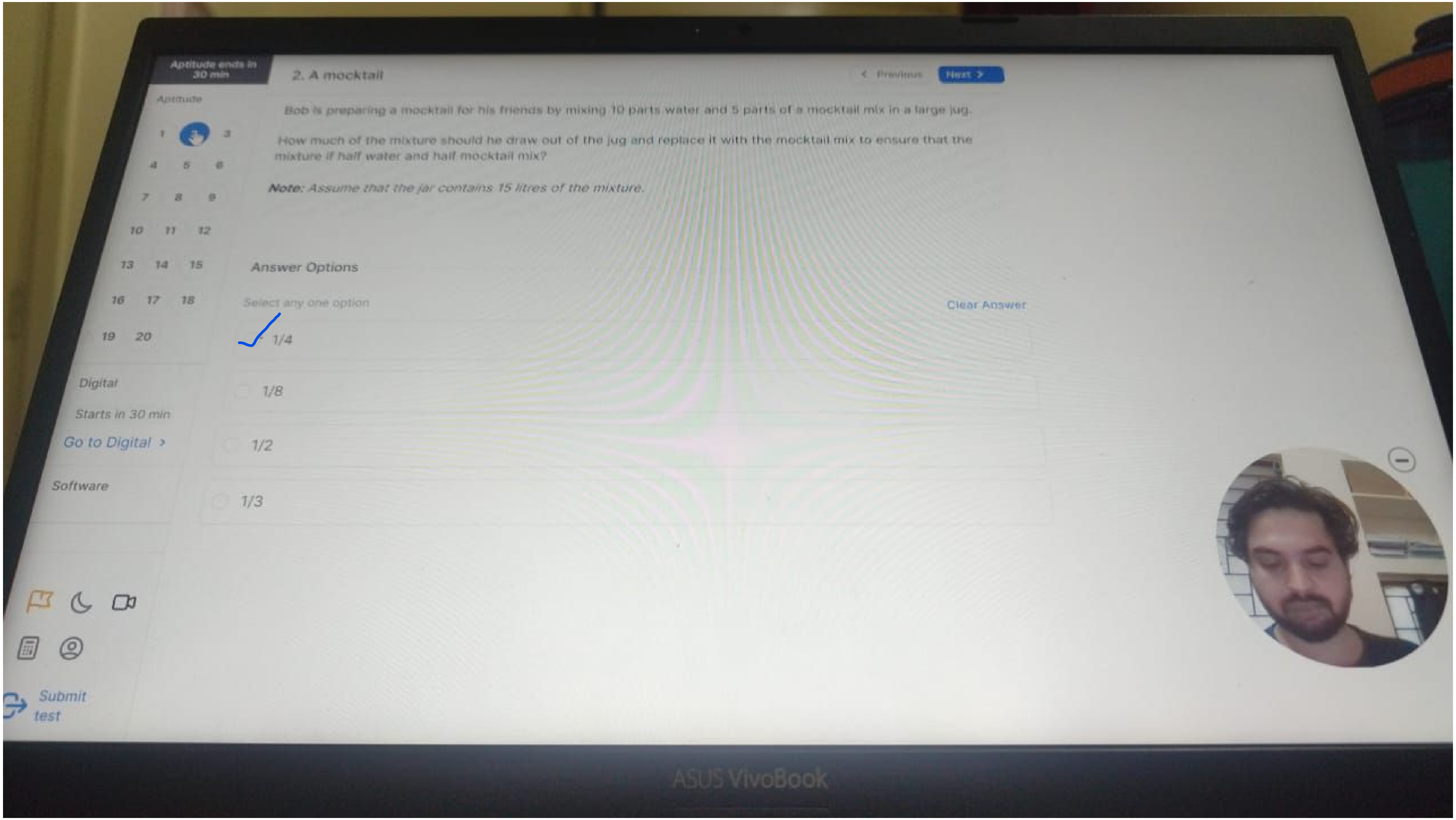

Aptitude ends in 30 min
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Digital
Starts in 30 min
Go to Digital ›
Software
Submit test
1. A circular plot
Next ›
A circular plot has been alloted by the government for the construction of a embassy building. The government decides to increase the area of the plot by 21%.
What would the percentage increase in the circumference of the plot be as a result of this?
Answer Options
Select any one option
Clear Answer
10%
15%
20%
11%
ASUS VivoBook


Aptitude ends in 30 min
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Digital
Starts in 30 min
Go to Digital >
Software
Submit test
2. A mocktail
< Previous
Next >
Bob is preparing a mocktail for his friends by mixing 10 parts water and 5 parts of a mocktail mix in a large jug.
How much of the mixture should he draw out of the jug and replace it with the mocktail mix to ensure that the mixture if half water and half mocktail mix?
Note: Assume that the jar contains 15 litres of the mixture.
Answer Options
Select any one option
Clear Answer
1/4
1/8
1/2
1/3
ASUS VivoBook

Aptitude ends in 30 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 30 min

Go to Digital >

Software

Submit test

3. A newly opened mall

< Previous Next >

A newly opened shopping mall has five elevators E1, E2, E3, E4 and E5. Elevators E1, E3 and E5 go to the even floors and elevators E2 and E4 go to the odd floors. All elevators go to the first and seventh floors.

Which of these choices must be **TRUE** in the given scenario?

Answer Options

Select any one option

Clear Answer

- A person that enters elevator E1 on the ground floor cannot reach the fifth floor.
- A person that enters the elevator E3 on the sixth floor cannot go to the ninth floor.
- A person must switch elevators to go from an even floor to an odd floor.
- None of the given options

ASUS VivoBook

Aptitude ends in 30 min

Aptitude

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Digital

Starts in 30 min

Go to Digital >

Software

Submit test

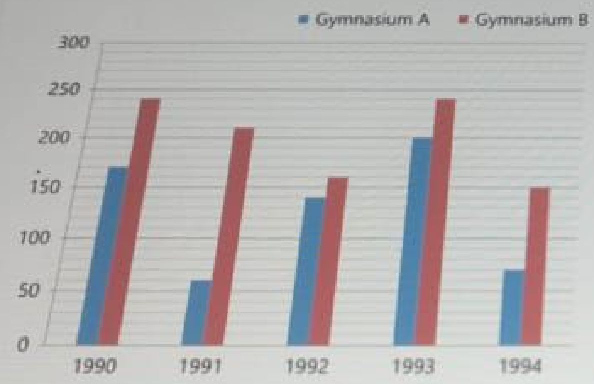
4. Gymnasiums A and B

< Previous Next >

The given bar graph shows the total number of members enrolled in gymnasiums A and B from 1990 to 1994.

For both A and B, it was observed that in the year 1995, there was a **30% increase** in the total number of members enrolled in comparison to the enrollment in 1994.

Find the total number of members enrolled in 1995.

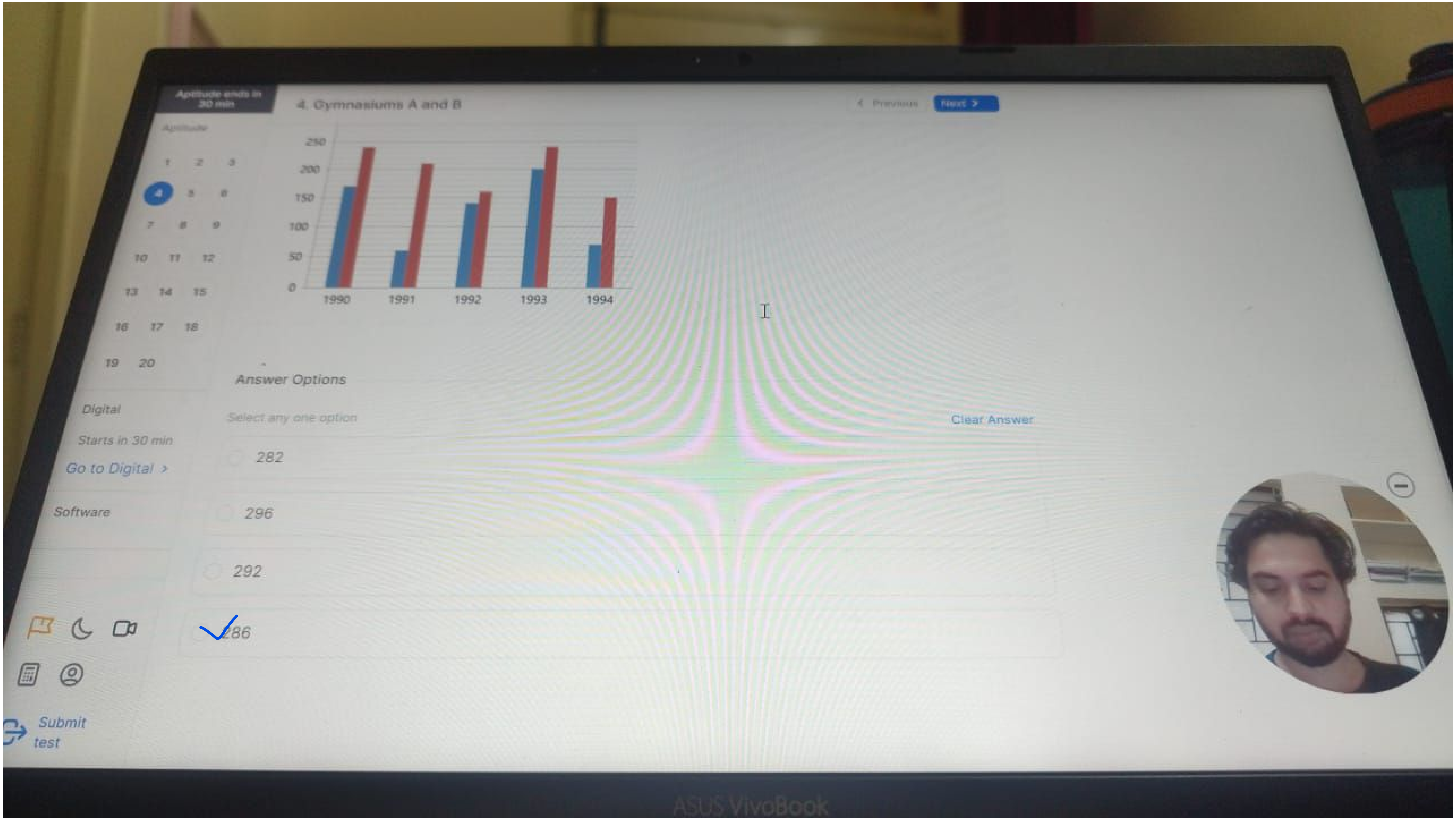


Answer Options

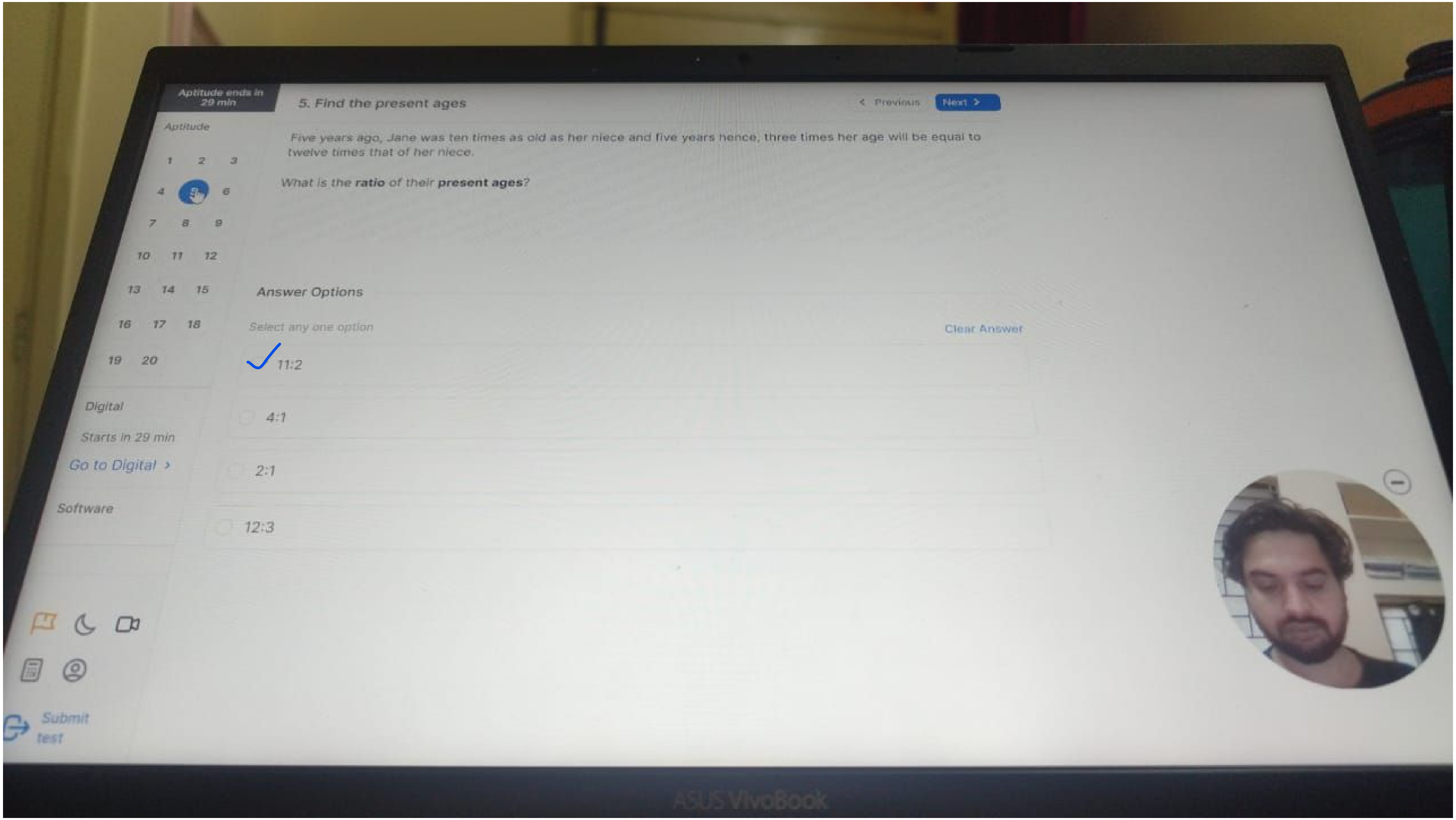
Select any one option

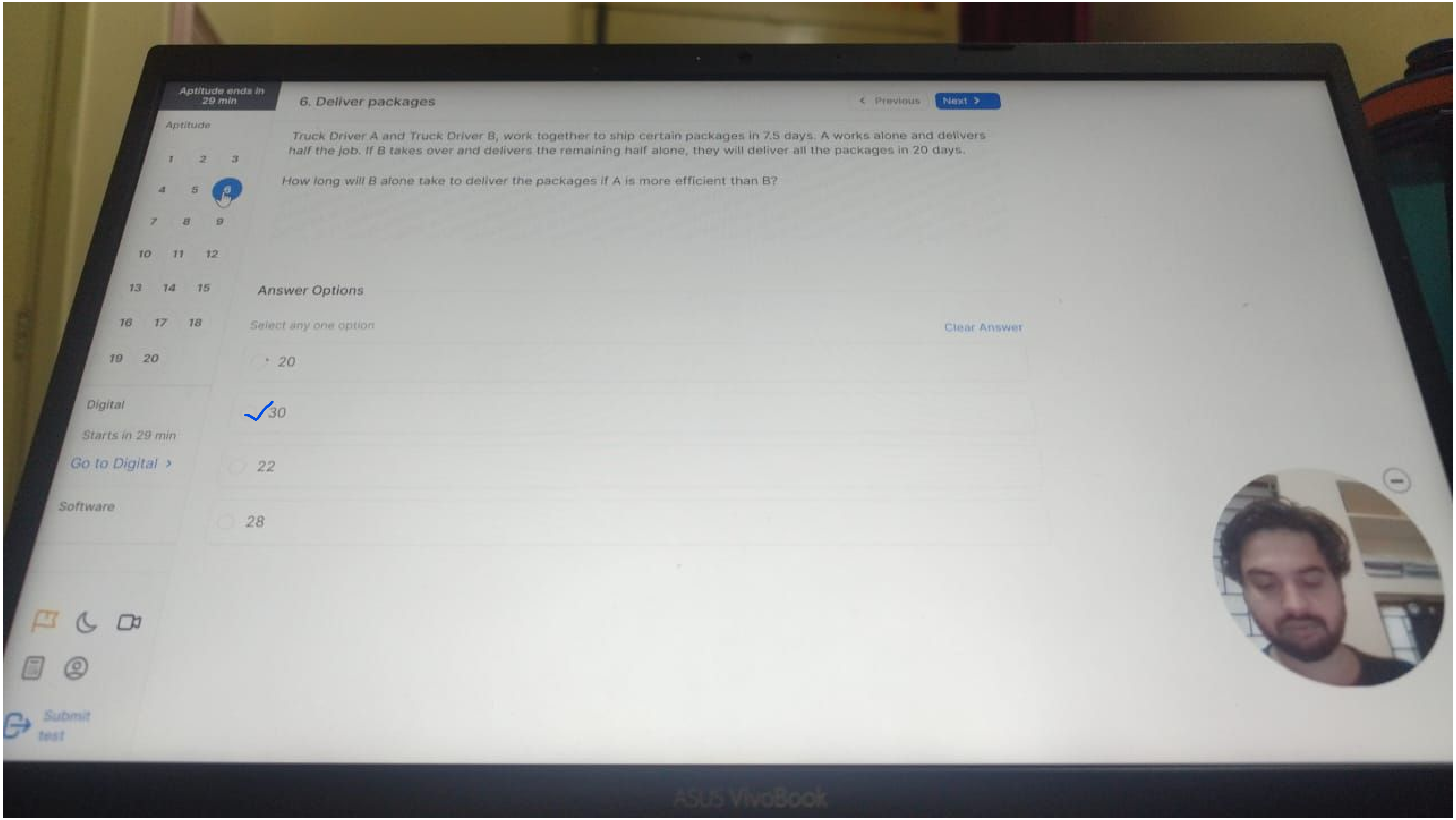
Clear Answer

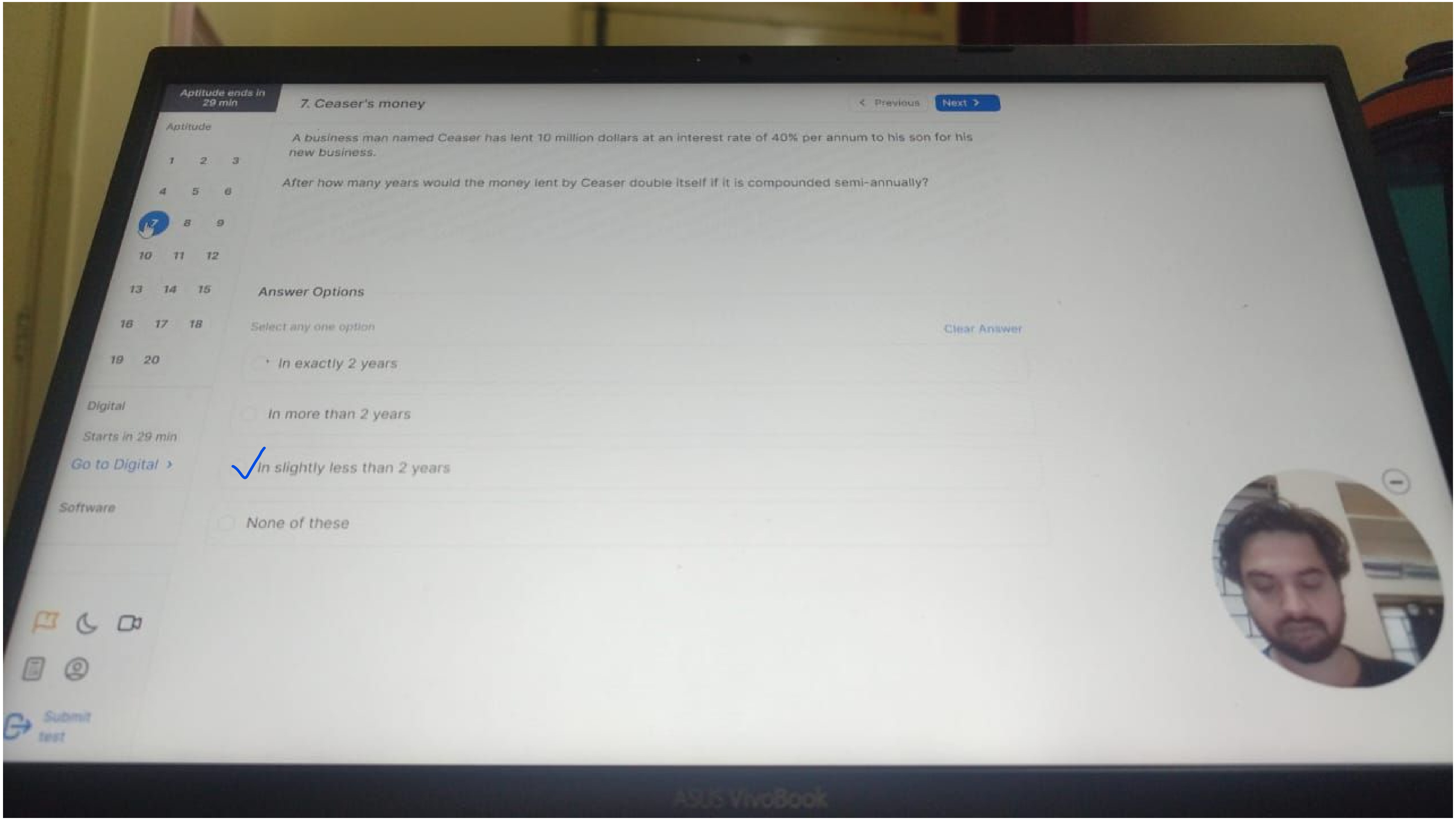
- 282
- 296

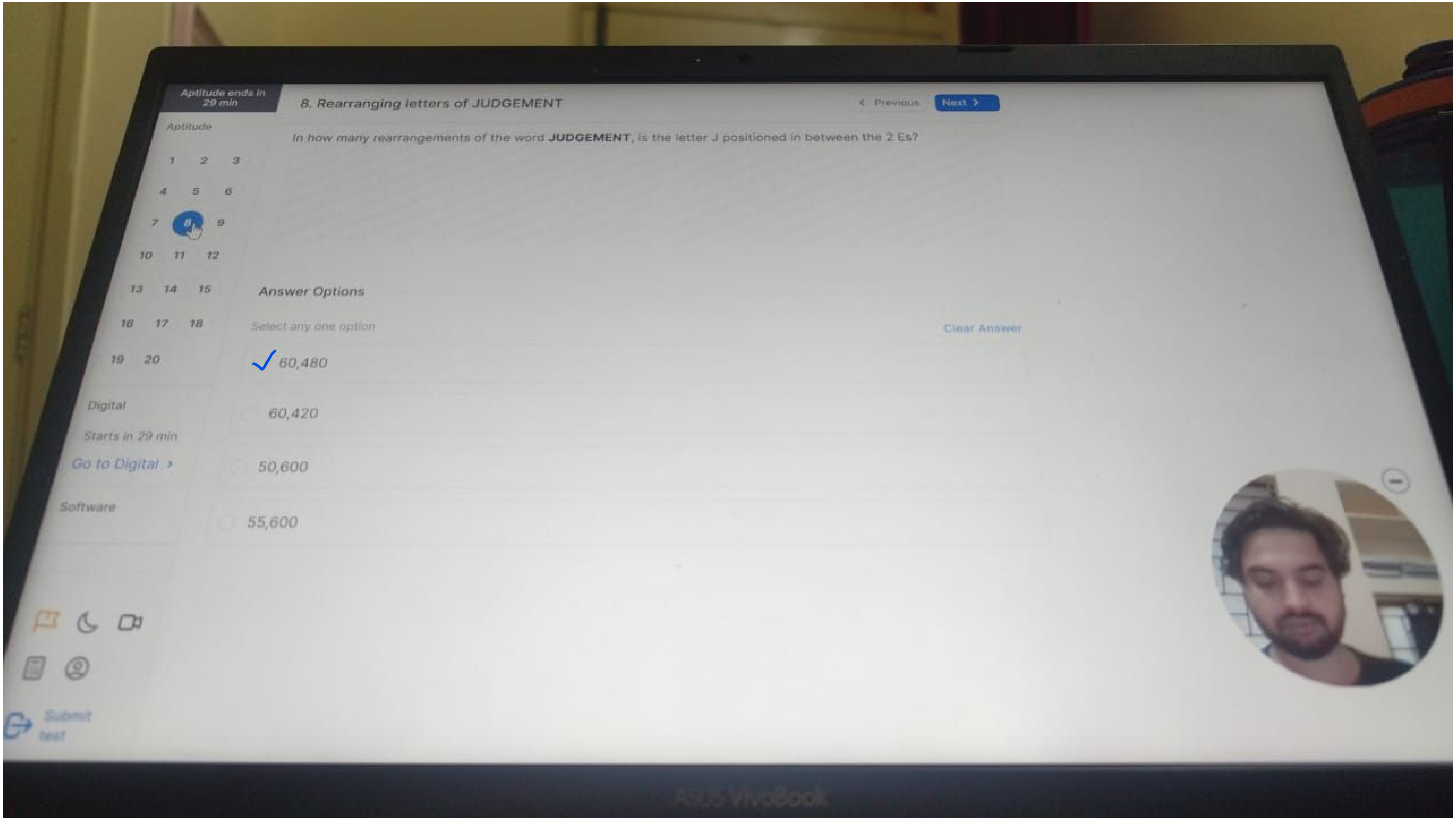


Aptitude ends in 30 min
4. Gymnasiums A and B
Previous
Next
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
250
200
150
100
50
0
1990
1991
1992
1993
1994
Answer Options
Select any one option
Clear Answer
282
296
292
286
Digital
Starts in 30 min
Go to Digital
Software
Submit test
ASUS VivoBook


Aptitude ends in 29 min
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Digital
Starts in 29 min
Go to Digital ›
Software
Submit test
5. Find the present ages
‹ Previous
Next ›
Five years ago, Jane was ten times as old as her niece and five years hence, three times her age will be equal to twelve times that of her niece.
What is the **ratio** of their **present ages**?
Answer Options
Select any one option
Clear Answer
11:2
4:1
2:1
12:3
ASUS VivoBook


Aptitude ends in 29 min
6. Deliver packages
Previous
Next
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Truck Driver A and Truck Driver B, work together to ship certain packages in 7.5 days. A works alone and delivers half the job. If B takes over and delivers the remaining half alone, they will deliver all the packages in 20 days.
How long will B alone take to deliver the packages if A is more efficient than B?
Answer Options
Select any one option
Clear Answer
20
30
22
28
Digital
Starts in 29 min
Go to Digital
Software
Submit test
ASUS VivoBook


Aptitude ends in 29 min
7. Ceaser's money
Previous
Next
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
A business man named Ceaser has lent 10 million dollars at an interest rate of 40% per annum to his son for his new business.
After how many years would the money lent by Ceaser double itself if it is compounded semi-annually?
Answer Options
Select any one option
Clear Answer
In exactly 2 years
In more than 2 years
In slightly less than 2 years
None of these
Digital
Starts in 29 min
Go to Digital
Software
Submit test
ASUS VivoBook


Aptitude ends in 29 min
8. Rearranging letters of JUDGEMENT
Previous
Next
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
In how many rearrangements of the word JUDGEMENT, is the letter J positioned in between the 2 Es?
Answer Options
Select any one option
Clear Answer
60,480
60,420
50,600
55,600
Digital
Starts in 29 min
Go to Digital
Software
Submit test
ASUS VivoBook

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 29 min

Go to Digital >

Software

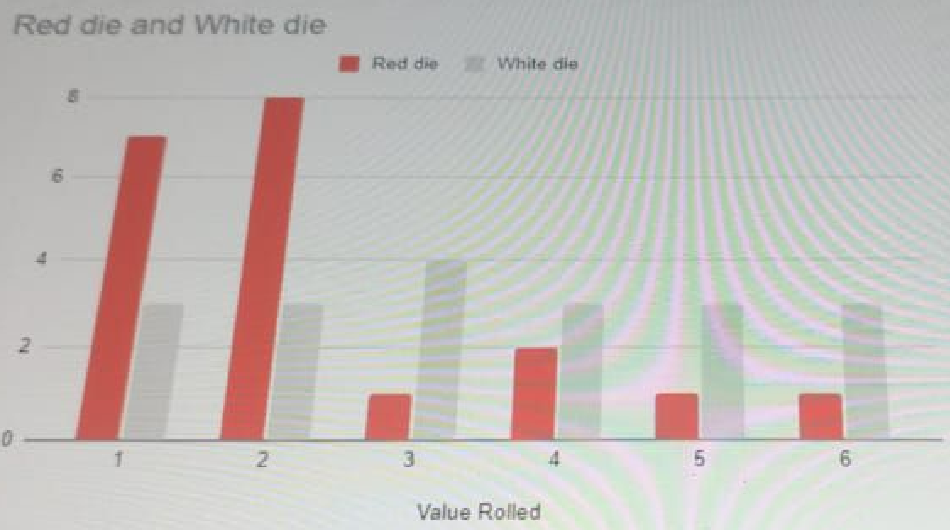
Submit test

9. A fair white die

< Previous Next >

The given bar graph shows the results from rolling one white and one red dice.

Assume that the white die is fair (has an equal chance of landing on each of its faces). What is the probability that the mean score will increase after one more roll?

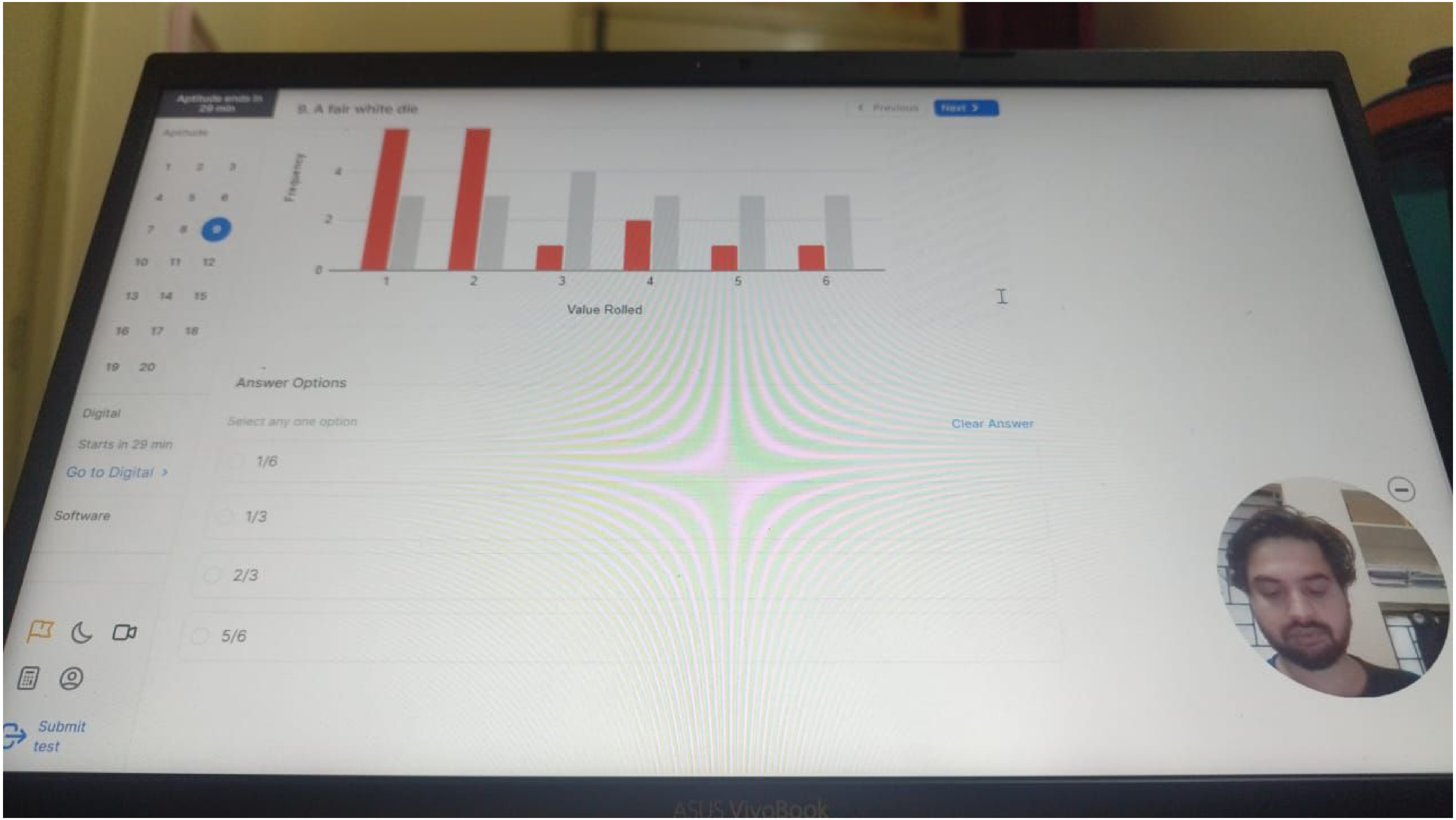


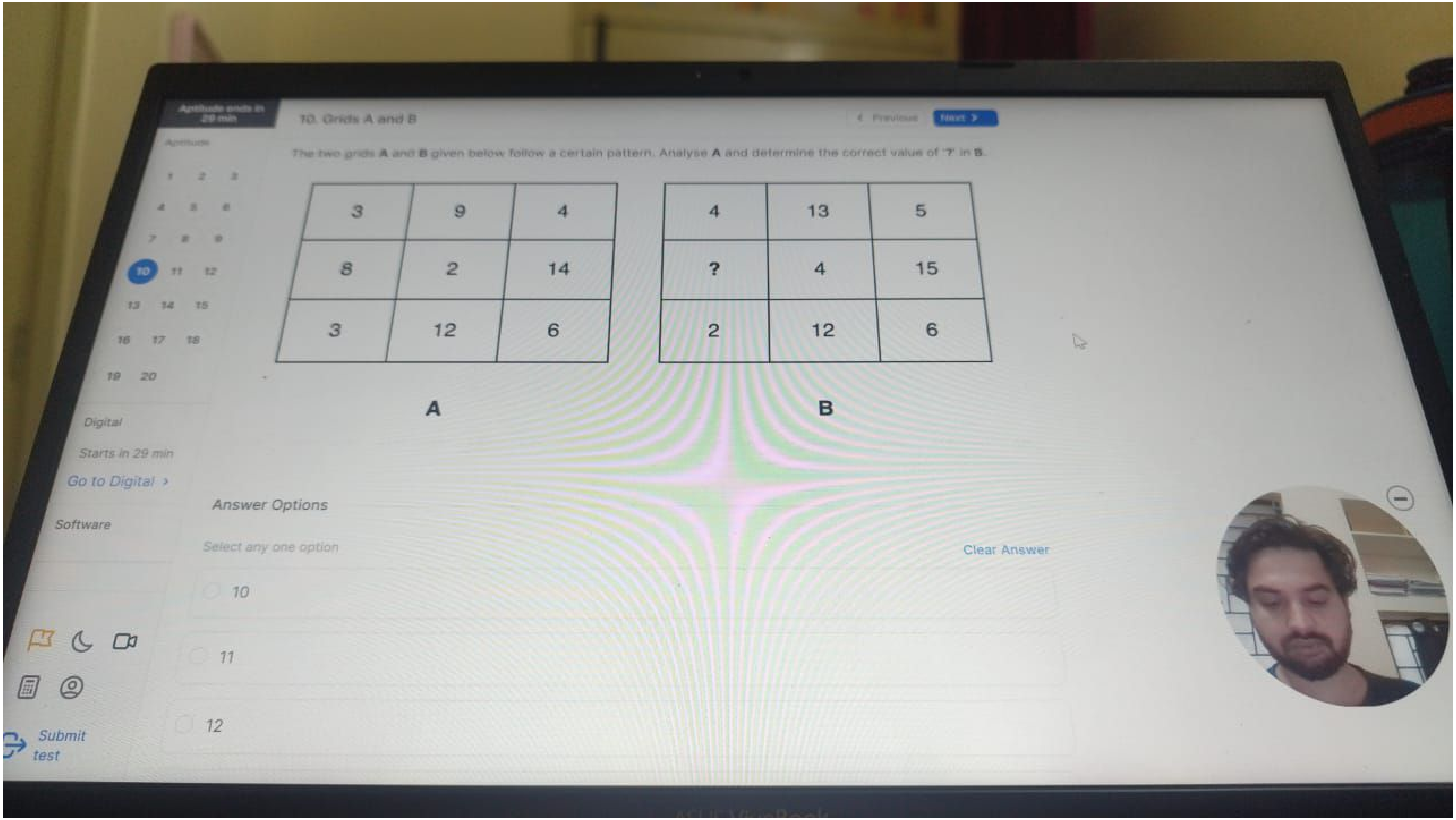
Answer Options

Select any one option

Clear Answer

1/6


Aptitude ends in 29 min
9. A fair white die
Previous
Next
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Frequency
4
2
0
1 2 3 4 5 6
Value Rolled
Answer Options
Select any one option
Clear Answer
1/6
1/3
2/3
5/6
Digital
Starts in 29 min
Go to Digital >
Software
Submit test
ASUS VivoBook



Aptitude ends in 29 min
10. Grids A and B
Previous
Next
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
The two grids A and B given below follow a certain pattern. Analyse A and determine the correct value of '?' in B.
3 | 9 | 4
8 | 2 | 14
3 | 12 | 6
A
4 | 13 | 5
? | 4 | 15
2 | 12 | 6
B
Digital
Starts in 29 min
Go to Digital
Software
Answer Options
Select any one option
Clear Answer
10
11
12
Submit test

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 29 min

Go to Digital >

Software

Submit test

11. Ratio of speeds of X and Y

< Previous   Next >

Two trains X and Y start at the same time from points A and B towards each other, and after crossing, they take 12 seconds and 3 seconds to reach B and A, respectively.

What is the ratio of the speed of X to Y?

Answer Options

Select any one option

Clear Answer

- 1:2
- 1:4
- 1:8
- Can not be determined

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 29 min

Go to Digital >

Software

Submit test

12. A luxury shoe store

< Previous Next >

A luxury shoe store wants to open a new franchise in a city. The store owner decides to install display units on all four walls of the store by hiring a fixture maker. However, on completion of the installation, the store owner complained saying that a portion of a high wall was missing display units.

The manager of the fixture company asked the workers as to whose fault it was but the workers refused to say anything. Hence, the manager decides to question the workers and the following facts are ascertained:

(1) No one other than W1, W2, W3 was involved in installing displays.
(2) W3 never does a job without the help of W1 (and possibly others) as an accomplice.
(3) W2 does not know how to install fixture on high walls.

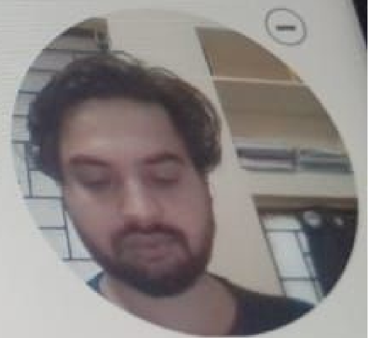
Is W1 innocent or guilty?

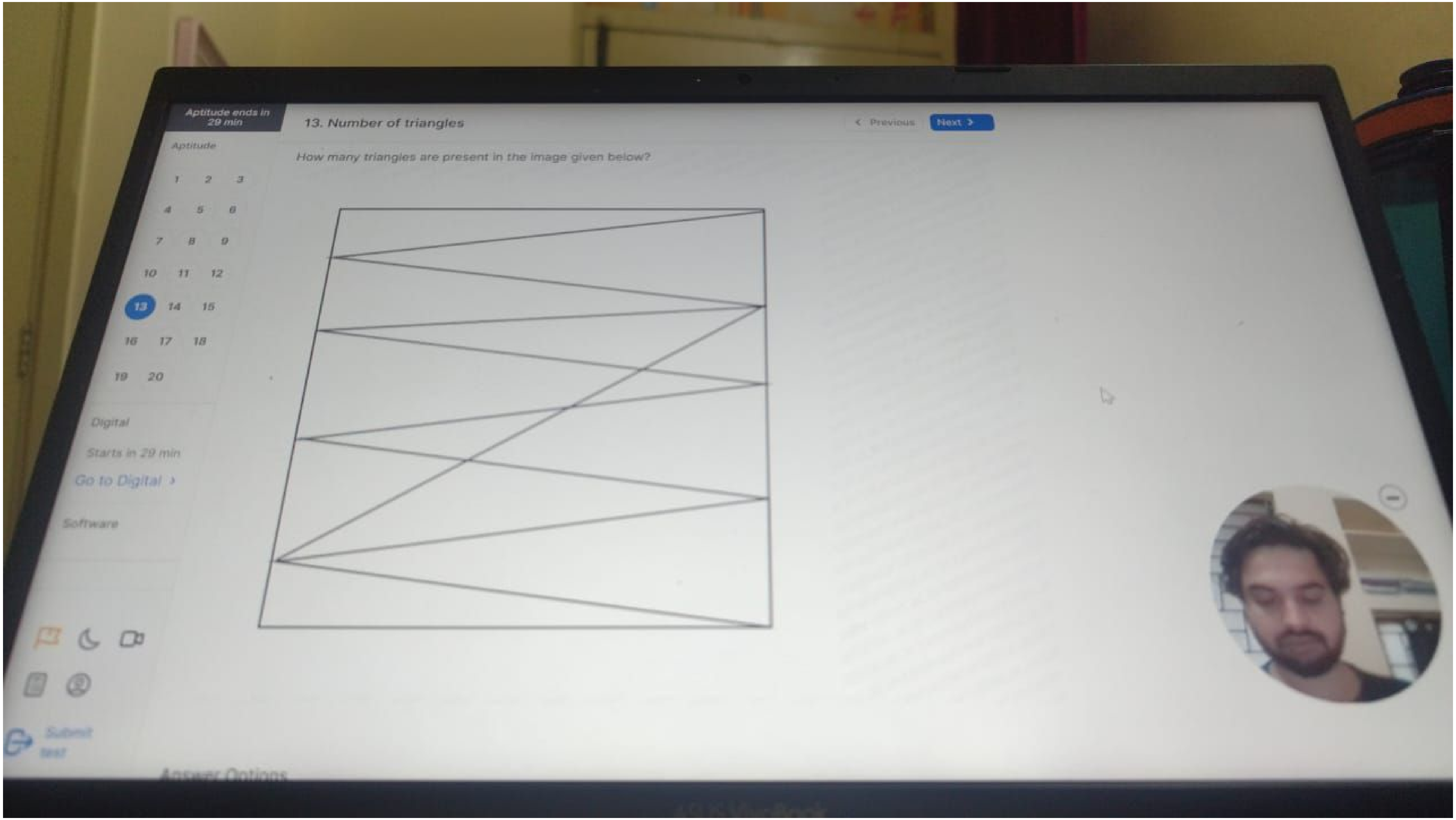
Answer Options

Select any one option

Clear Answer

- No, W1 is innocent.
- Yes, W1 is guilty.
- ✓ Insufficient information
- Cannot be determined

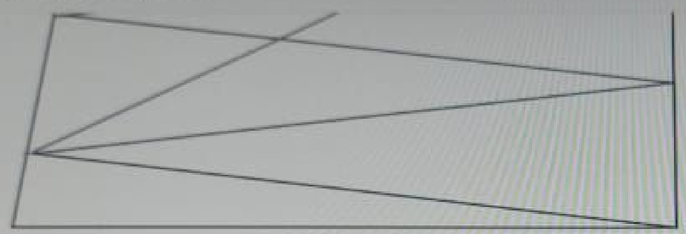



Aptitude ends in 29 min
13. Number of triangles
Previous
Next
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
How many triangles are present in the image given below?
Digital
Starts in 29 min
Go to Digital ›
Software
Submit test
Answer Options

Aptitude ends in 28 min

13. Number of triangles

< Previous | Next >



Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 28 min

Go to Digital >

Software

Answer Options

Select any one option

Clear Answer

- 20
- 22
- 23
- ✓ 25

Submit test

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Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 28 min

Go to Digital ›

Software

Submit test

15. New clock in a library

‹ Previous Next ›

On a particular day, a library installed a new clock which gains time uniformly. The librarian observes the clock on a Monday morning and notices that the clock is 10 mins slow at 10:00 AM.

She then observes the clock again on the following Monday and notices that the clock is 10 mins 50 seconds fast at 10:00 PM.

Analyse the given scenario and determine when this clock will show the correct time.

Note: *Select the accurate time in hours and minutes.*

Answer Options

Select any one option

Clear Answer

- 12:25 pm on Thursday
- 12:25 am on Thursday
- 10:00 pm on Thursday
- 10:25 pm on Wednesday

Aptitude ends in 28 min

14. Assorted coffee box

‹ Previous   Next ›

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

A newly launched coffee company wants to conduct a survey for its assorted pack containing three flavours hazelnut, Irish cream and caramel. To do this, 40 people were asked to rank the three flavours from 1 - 3, 1 being the most preferred and 3 being the least preferred.

All 40 responses were gathered and it was observed that no two flavours were ranked equally by any people that attended the survey. It was also observed that 1/5 people ranked Irish cream last, 2/5 people ranked Irish cream before hazelnut and 1/4 people ranked Irish cream before caramel.

Analyse the given scenario and determine the number of people that ranked **Irish cream first**.

Answer Options

Select any one option

Clear Answer

✓ 6   check

5

2

4

Digital

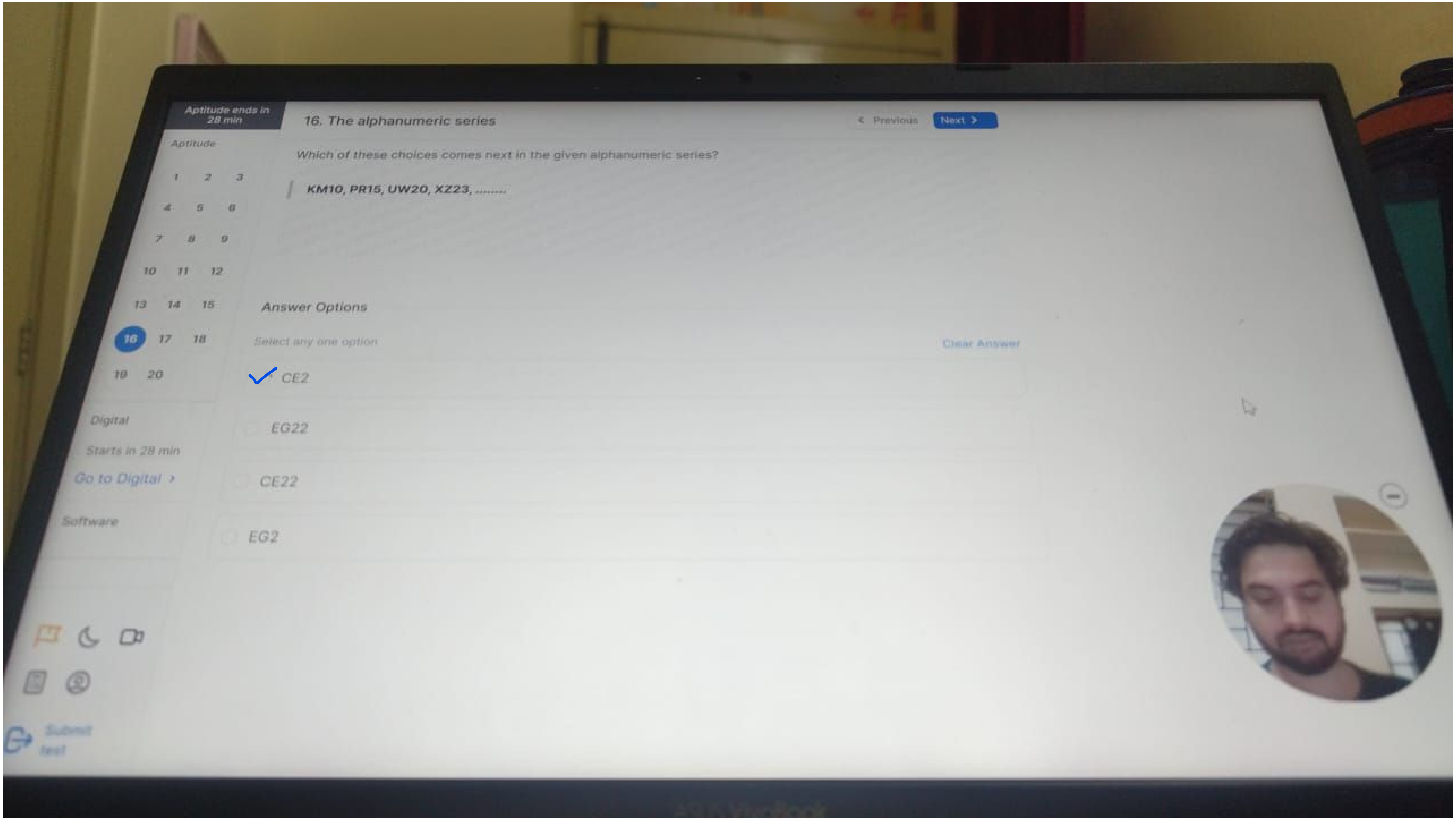
Starts in 28 min

Go to Digital ›

Software

Submit test

ASUS VivoBook


Aptitude ends in 28 min
16. The alphanumeric series
Previous
Next
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Which of these choices comes next in the given alphanumeric series?
KM10, PR15, UW20, XZ23, ...
Answer Options
Select any one option
Clear Answer
CE2
EG22
CE22
EG2
Digital
Starts in 28 min
Go to Digital
Software
Submit test

Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 28 min

Go to Digital ›

Software

Submit test

17. Disappointed customers

‹ Previous Next ›

Jake is a butcher. Given below is his selling procedure:

1. Get the order from the customer
2. Weigh the meat on the goods pan against the weights put on the weighing pan
3. Give it to the customer.

However, certain criteria need to be met.

1. The order can be anything in kilograms as long as it is a whole number.
2. The weight cannot be placed in the goods pan in the weighing machine.
3. Jake has to weigh the meat at one go and can't-do it in parts.
4. If Jake can't measure the weight, he sends them back.

Jake has weights of 1, 2, 4, 8, 16, 32, 64 kilograms. Customers start coming to Jake's shop right from the morning. A customer buys the same weight of meat as his number. For instance, customer number 32 buys 32 kilograms of meat. A total of 128 customers come to Jake's shop on Wednesday.

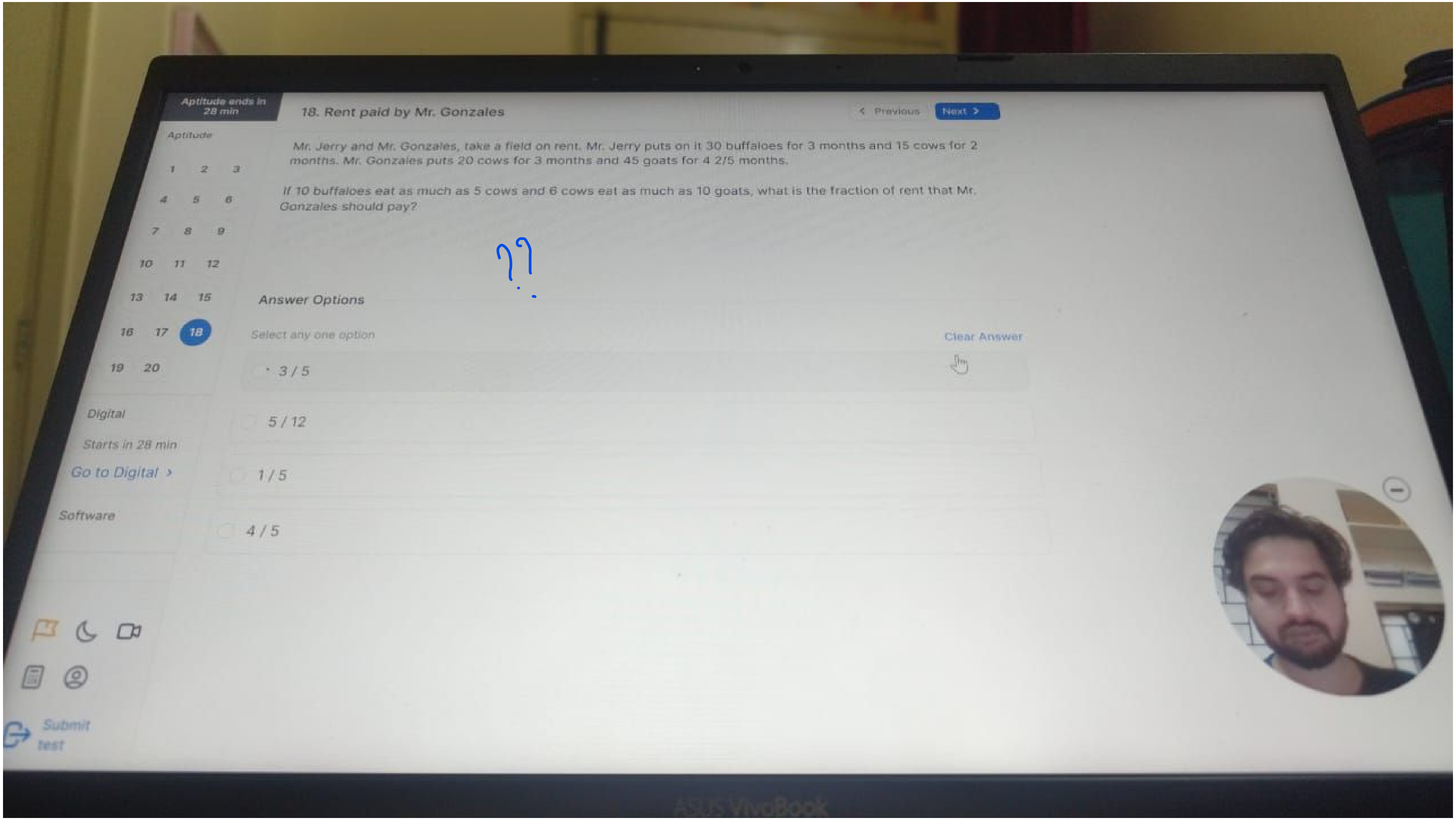
How many customers went away disappointed that day?

Answer Options

Select any one option

Clear Answer

- 0
- ✓ 1
- 2
- 3


Aptitude ends in 28 min
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Digital
Starts in 28 min
Go to Digital ›
Software
Submit test
18. Rent paid by Mr. Gonzales
‹ Previous
Next ›
Mr. Jerry and Mr. Gonzales, take a field on rent. Mr. Jerry puts on it 30 buffaloes for 3 months and 15 cows for 2 months. Mr. Gonzales puts 20 cows for 3 months and 45 goats for 4 2/5 months.
If 10 buffaloes eat as much as 5 cows and 6 cows eat as much as 10 goats, what is the fraction of rent that Mr. Gonzales should pay?
??
Answer Options
Select any one option
Clear Answer
3 / 5
5 / 12
1 / 5
4 / 5
ASUS VivoBook

Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 28 min

Go to Digital ›

Software

Submit test

19. Plant enthusiasts

‹ Previous   Next ›

Two house plant enthusiasts - John and Ben have a wide range of succulents in the ratio 8:15, respectively. John and Ben meet at a plant collective one day and decide to exchange some succulents. Ben gives some of his succulents to Ben which changes the original ratio to 6:11.

Analyse the scenario and determine the fraction of succulents that Ben gave to John from his original collection.

Answer Options

Select any one option   ??   Clear Answer

- 2/75
- 1/50
- 3/15
- 2/5

ASUS VivoBook

Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 28 min

Go to Digital ›

Software

Submit test

20. When will the tank overflow

‹ Previous   Next ›

Pipe A can fill a tank in 10 hours, pipe B can fill it in 20 hours, and an outlet pipe, C, can empty the tank in 30 hours. Each tap is opened, one by one, for exactly one hour and then closed.

If the tank is 1/4th full, and the pipes start working in alphabetic order (A, B, and then C), after how long will the tank begin to overflow?

Answer Options

Select any one option   Clear Answer

- 18hrs 30 mins
- 15hrs 30 mins
- 18hrs 20 mins
- 17hrs 30 mins

ASUS VivoBook

Digital ends in 45 min

Aptitude

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Software

Starts in 45 min

Go to Software ›

1. The minimized representation of a function

< Previous Next >

It is given that $f(A,B,C,D) = \Pi M\,(0,1,3,4,5,7,9,11,12,13,14,15)$ is a max-term representation of a Boolean function **f(A,B,C,D)**.

What will be the equivalent minimized representation of this function?

Note: *A is the MSB and D is the LSB.*

Answer Options

Select any one option

Clear Answer

- $(A+C'+D)\,(A'+B+D)$
- $AC'D + A'BD$
- $A'CD'+AB'CD'+AB'C'D'$
- $(B+C'+D)\,(A+B'+C'+D)\,(A'+B+C+D)$

Submit test

ASUS VivoBook

Digital ends in 45 min

Aptitude

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Software

Starts in 45 min

Go to

Submit test

2. A network with multiple branches and nodes

< Previous Next >

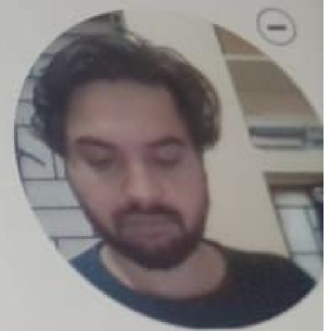
In a network with multiple branches and nodes, which of the following statements regarding Kirchhoff's Laws is correct?

Answer Options

Select any one option

Clear Answer

- Kirchhoff's Voltage Law (KVL) is used to analyze current flow at nodes in a circuit.
- Kirchhoff's Current Law (KCL) is used to analyze voltage drops across components in a circuit.
- Kirchhoff's Voltage Law (KVL) states that the algebraic sum of currents entering and leaving a node is zero.
- Kirchhoff's Current Law (KCL) states that the algebraic sum of voltages around any closed loop in a circuit i s zero.



ASUS VivoBook

Digital ends in 44 min

Aptitude

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Software

Starts in 44 min

Go to

Submit test

3. A memory unit with 8 data pins

< Previous | Next >

A memory unit has 8 data pins and 16 address lines.

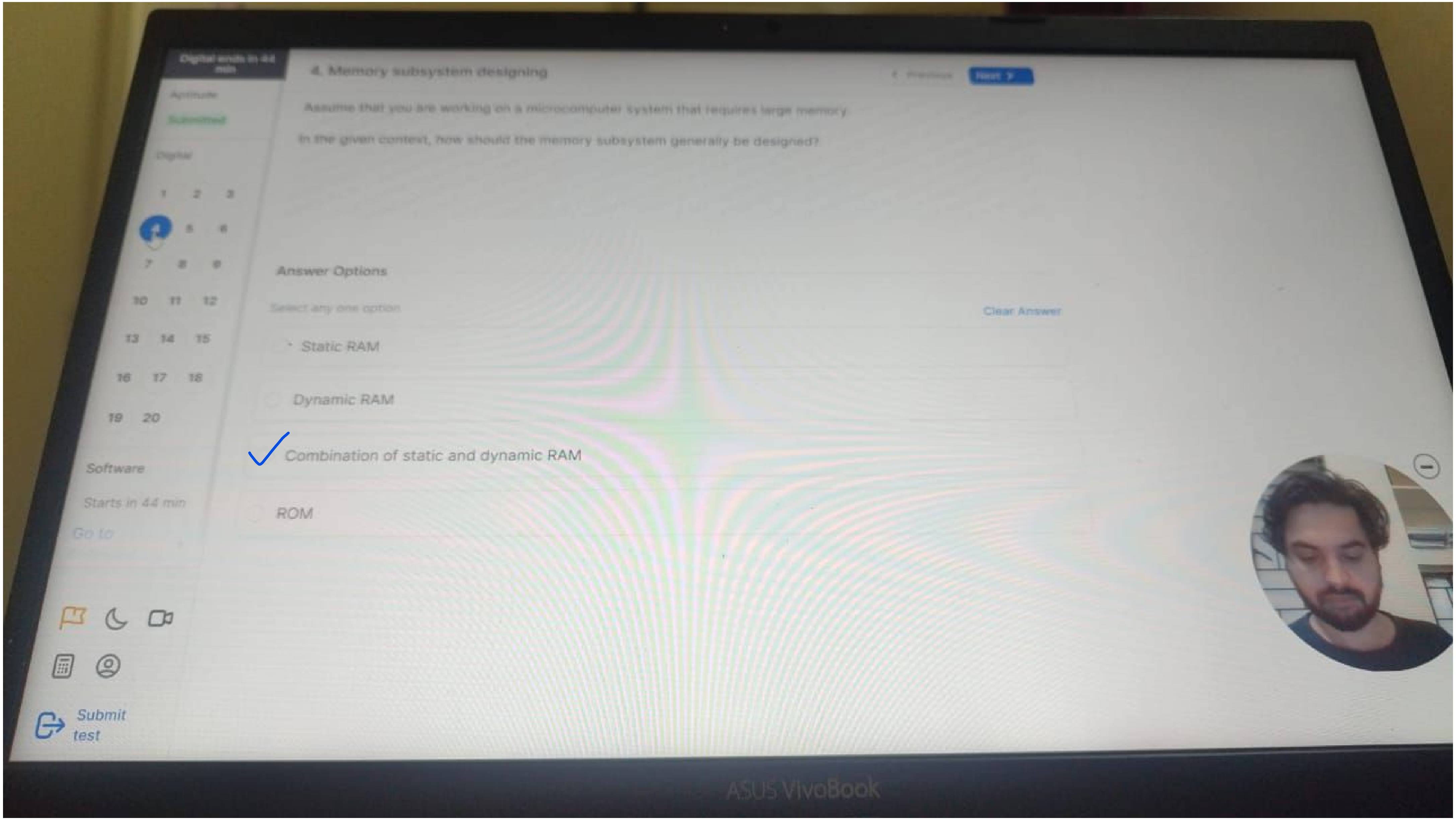
What will be its total capacity?

Answer Options

Select any one option

Clear Answer

- 128 KB
- 256 KB
- 512 KB
- 1024 KB
- None of these


Digital ends in 44 min
Aptitude
Submitted
Digital
4. Memory subsystem designing
Next
Assume that you are working on a microcomputer system that requires large memory
In the given context, how should the memory subsystem generally be designed?
1 2 3
5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Answer Options
Select any one option
Clear Answer
Static RAM
Dynamic RAM
Combination of static and dynamic RAM
ROM
Software
Starts in 44 min
Go to
Submit test
ASUS VivoBook

Digital ends in 44 min

Aptitude

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Software

Starts in 44 min

Go to

Submit test

5. Declaring a function pointer

< Previous   Next >

Consider the following C code

```
#include <stdio.h>
void fun(int a)
{
printf("Value of a is %d\n", a);
}
int main()
{
XXX
(*fun_ptr)(10);
return 0;
}
```

What should you use in place of XXX to declare a function pointer *fun_ptr to fun()?

Answer Options

Select any one option

Clear Answer

- void (*fun_ptr)(int) = &fun;
- void (*fun_ptr)(int);
  fun_ptr = &fun;
- ✓ Both Choice 1 and 2
- None of these

ASUS VivoBook

Digital ends in 44 min

Aptitude

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Software

Starts in 44 min

Go to

Submit test

6. Using a CMOS inverter

< Previous Next >

You wish to use a CMOS inverter as an amplifier.

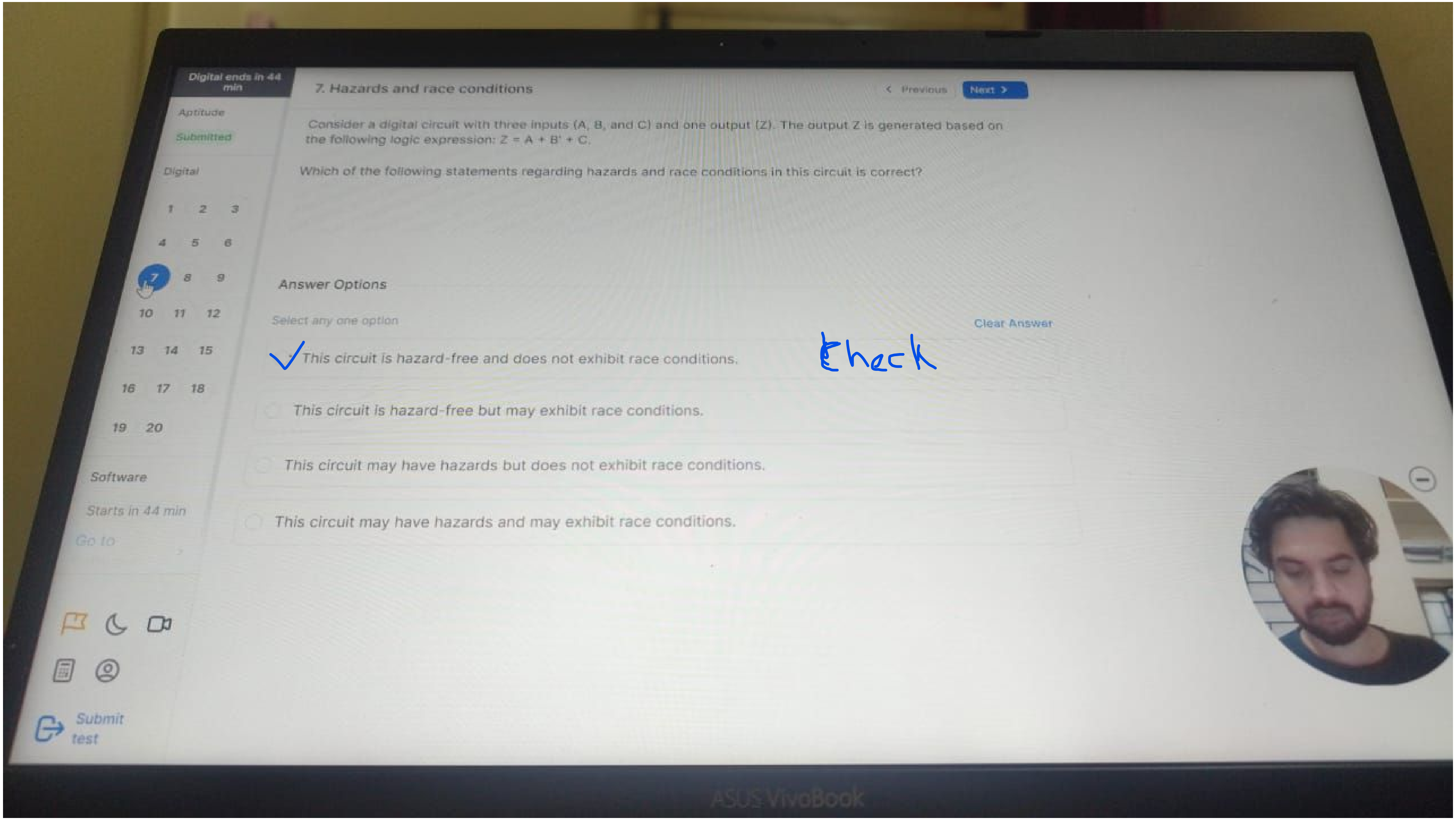
Which of the given conditions need to be **true** in the above scenario?

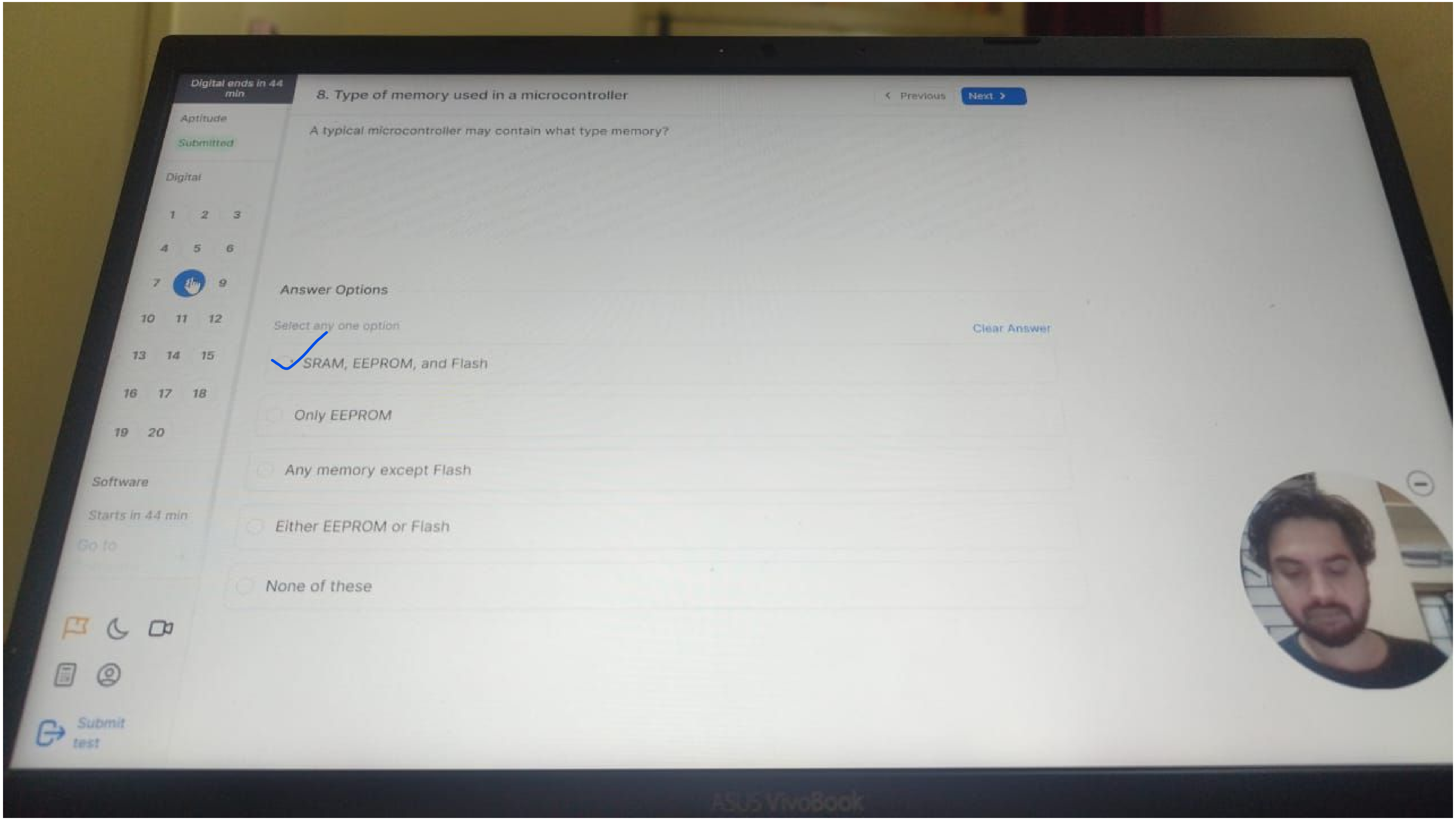
Answer Options

Select any one option

Clear Answer

- PMOS should be in linear region and NMOS in cut-off region
- Both PMOS and NMOS should be in linear region.
- ✓ Both PMOS and NMOS should be in saturation region.
- NMOS should be in linear region and PMOS in cut-off region


Digital ends in 44 min
Aptitude
Submitted
Digital
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Software
Starts in 44 min
Go to
Submit test
7. Hazards and race conditions
Previous
Next
Consider a digital circuit with three inputs (A, B, and C) and one output (Z). The output Z is generated based on the following logic expression: Z = A + B' + C.
Which of the following statements regarding hazards and race conditions in this circuit is correct?
Answer Options
Select any one option
Clear Answer
This circuit is hazard-free and does not exhibit race conditions.
Check
This circuit is hazard-free but may exhibit race conditions.
This circuit may have hazards but does not exhibit race conditions.
This circuit may have hazards and may exhibit race conditions.
ASUS VivoBook


Digital ends in 44 min
Aptitude
Submitted
Digital
1 2 3
4 5 6
7 9
10 11 12
13 14 15
16 17 18
19 20
Software
Starts in 44 min
Go to
Submit test
8. Type of memory used in a microcontroller
Previous
Next
A typical microcontroller may contain what type memory?
Answer Options
Select any one option
Clear Answer
SRAM, EEPROM, and Flash
Only EEPROM
Any memory except Flash
Either EEPROM or Flash
None of these
ASUS VivoBook

Digital ends in 44 min

Aptitude

Submitted

Digital

1 2 3

4 5 6

7 8

10 11 12

13 14 15

16 17 18

19 20

Software

Starts in 44 min

Go to

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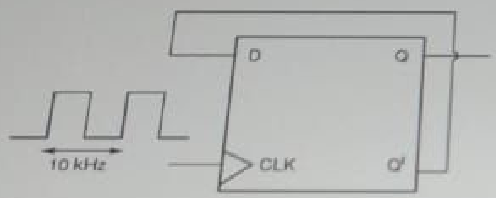
9. The frequency of the output signal

< Previous Next >

Assume that a clock signal of frequency 10 kHz is applied to the rising edge-triggered D flip-flop shown in figure.

In the given context, what will be the frequency of the output signal available at **Q**?

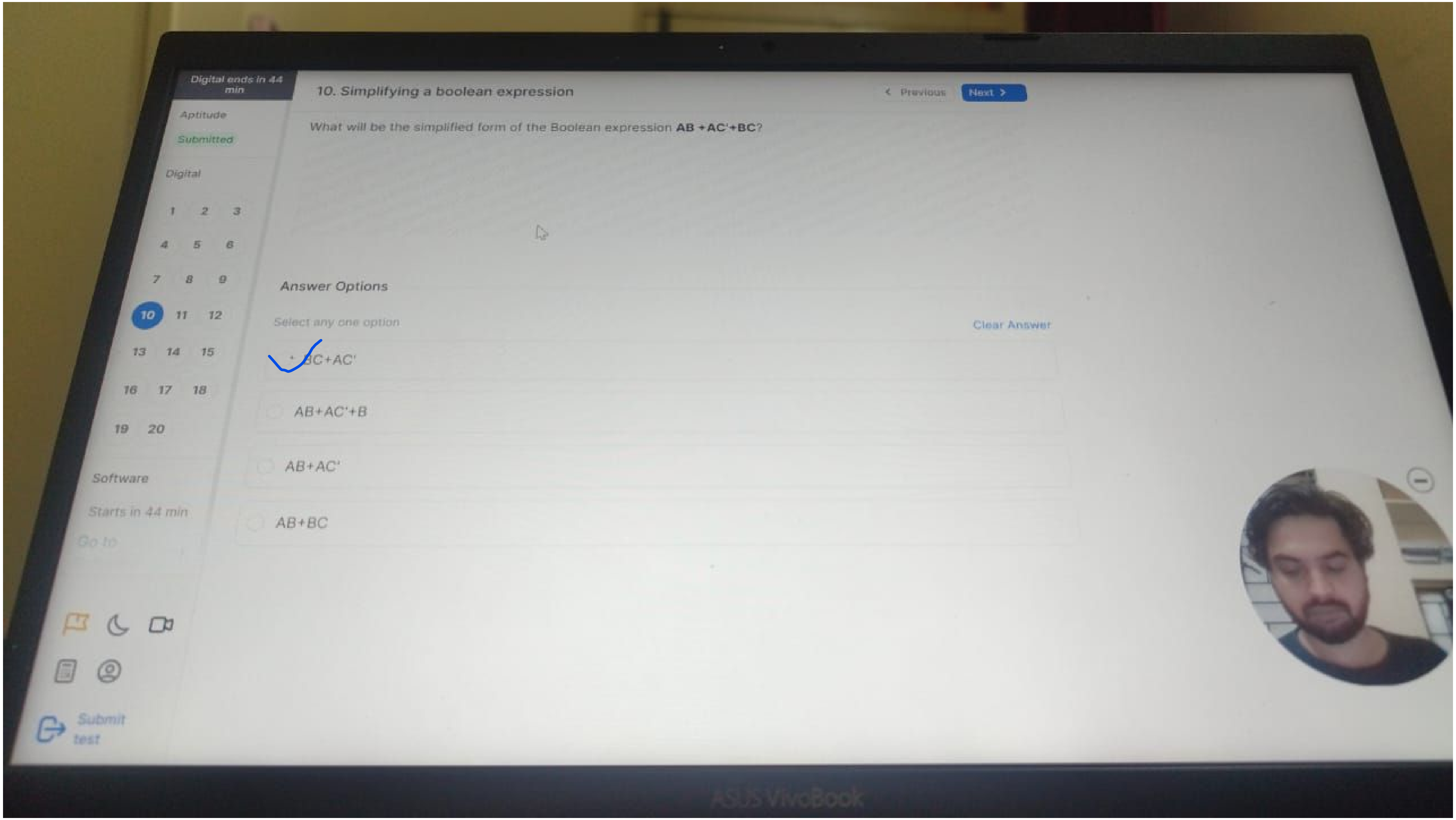


Answer Options

Select any one option

Clear Answer

- 10 kHz
- 20 kHz
- 2.5 kHz
- 5 kHz


Digital ends in 44 min
Aptitude
Submitted
Digital
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Software
Starts in 44 min
Go to
Submit test
10. Simplifying a boolean expression
Previous
Next
What will be the simplified form of the Boolean expression AB +AC'+BC?
Answer Options
Select any one option
Clear Answer
BC+AC'
AB+AC'+B
AB+AC'
AB+BC
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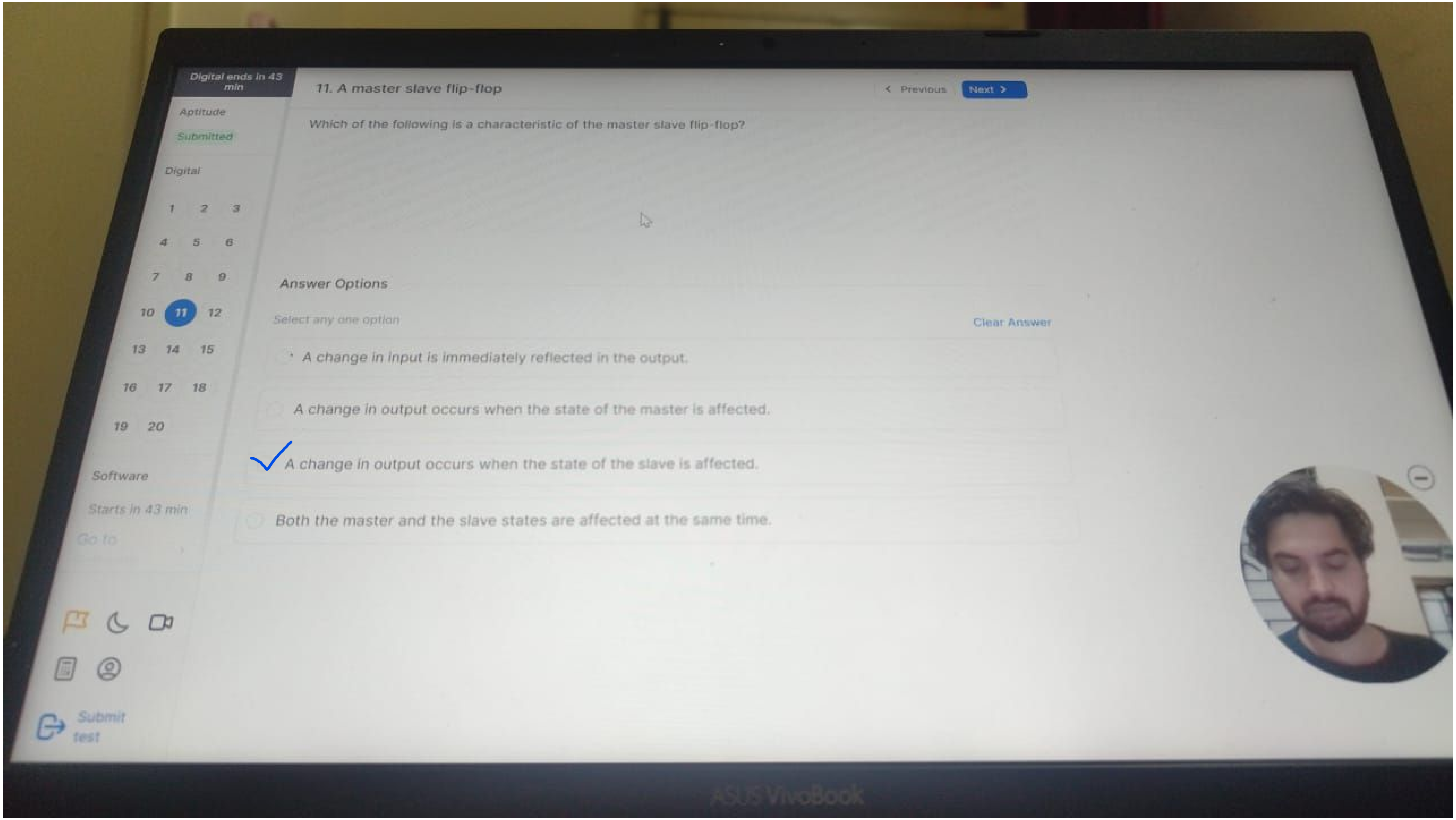
Digital ends in 43 min
Aptitude
Submitted
Digital
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Software
Starts in 43 min
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Submit test
11. A master slave flip-flop
Previous
Next
Which of the following is a characteristic of the master slave flip-flop?
Answer Options
Select any one option
Clear Answer
A change in input is immediately reflected in the output.
A change in output occurs when the state of the master is affected.
A change in output occurs when the state of the slave is affected.
Both the master and the slave states are affected at the same time.
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Digital ends in 43 min

Aptitude

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Software

Starts in 43 min

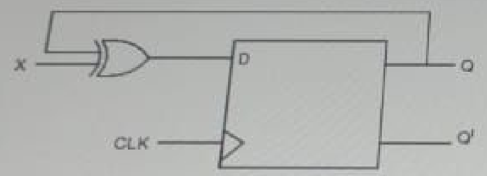
Go to

Submit test

12. Representing digital circuits

< Previous Next >

Which of the following represents the digital circuit given below?



Answer Options

Select any one option

Clear Answer

- J-K flip-flop
- Clocked RS flip-flop
- T flip-flop
- Ring Counter

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Digital ends in 43 min

Aptitude

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

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16 17 18

19 20

Software

Starts in 43 min

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13. Response of a system

‹ Previous Next ›

A control system is defined by the given mathematical relationship:

> d^2x/dt^2 + 6. dx/dt+ 5x = 12(1-e^-2t), where x(t=0) = 0 and x'(t=0) = 0

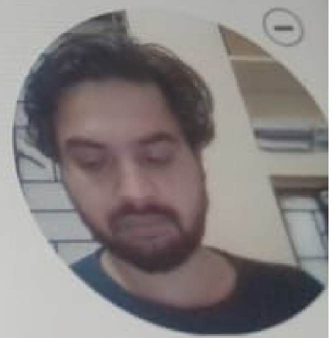
Calculate the response of the system as $t \to \infty$.

Answer Options

Select any one option

Clear Answer

- x = 6
- x = 2
- x =2.4
- x = -2



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Digital ends in 43 min

Aptitude

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Software

Starts in 43 min

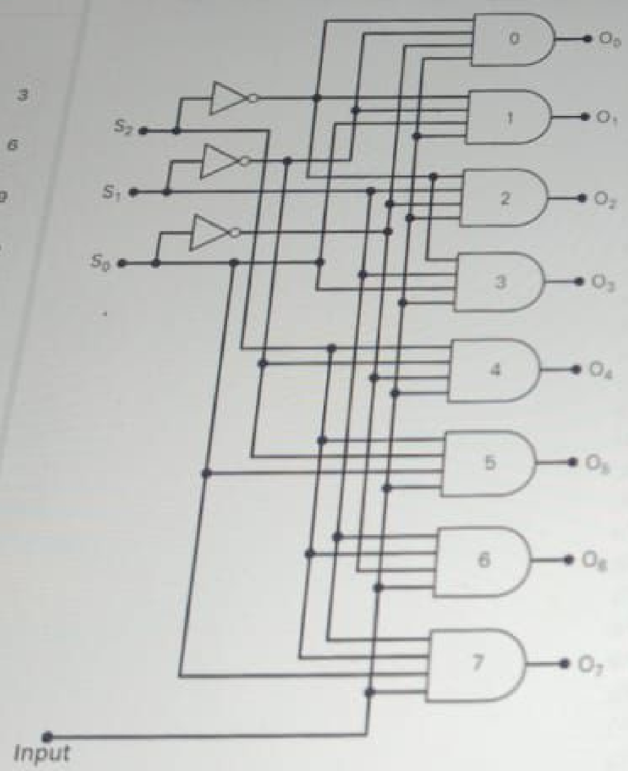
Go to

Submit test

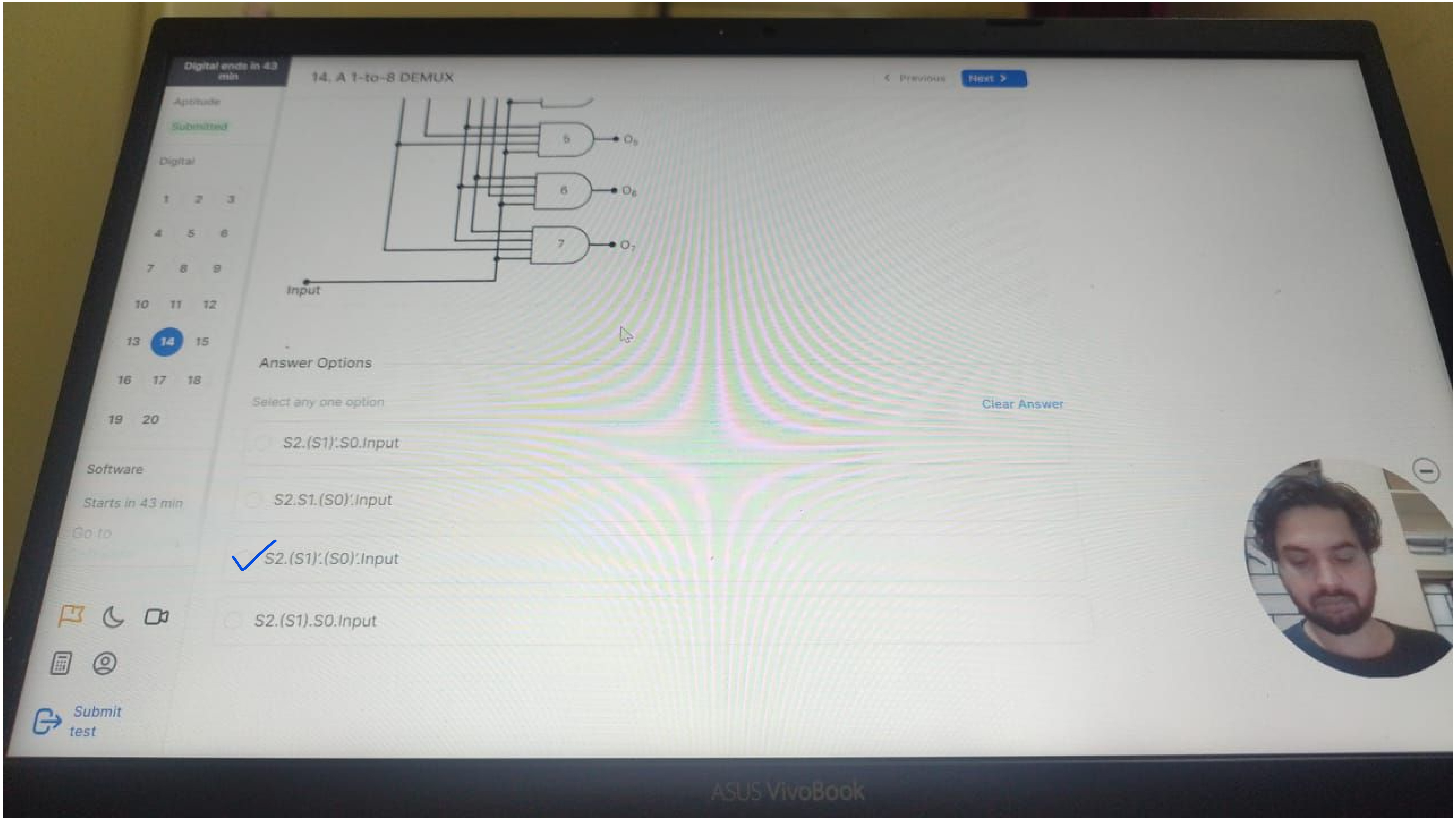
14. A 1-to-8 DEMUX

‹ Previous Next ›

What is the output **O4** of the following 1-to-8 demultiplexer?



Answer Options



Digital ends in 43 min
Aptitude
Submitted
Digital
1 2 3
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10 11 12
13 14 15
16 17 18
19 20
Software
Starts in 43 min
Go to
Submit test
14. A 1-to-8 DEMUX
Previous
Next
5
O5
6
O6
7
O7
Input
Answer Options
Select any one option
Clear Answer
S2.(S1)'.S0.Input
S2.S1.(S0)'.Input
S2.(S1)'.(S0)'.Input
S2.(S1).S0.Input
ASUS VivoBook

Digital ends in 43 min

Aptitude

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Software

Starts in 43 min

Go to

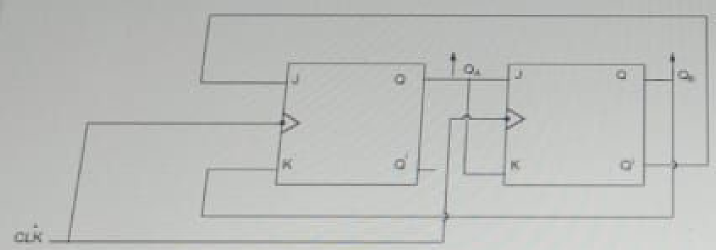
Submit test

15. The state of the counter

< Previous   Next >

A 2-bit counter circuit is shown below. It is given that the state of the counter at clock time **tn** is '10'.

What will be the state of the counter (**QA, QB**) after three cycles?

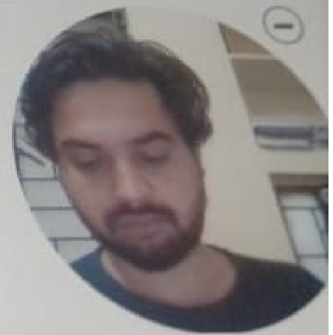


Answer Options

Select any one option

Clear Answer

- 0
- 1
- 10
- 11



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Digital ends in 43 min

Aptitude

Submitted

Digital

1 2 3
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Software

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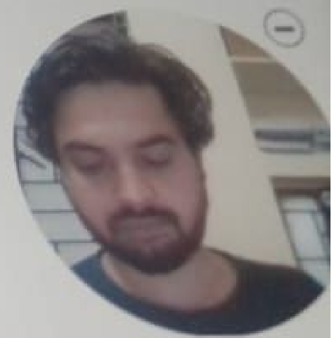
16. Working with arrays

< Previous Next >

Predict the output of the following C code snippet.

```
bool compare (const void * a, const void * b)
{
return ( *(int*)a == *(int*)b ); //L1
}
int search(void *arr, int arr_size, int ele_size, void *x,
bool compare (const void * , const void *)) //L2
{
char *ptr = (char *)arr;
int i;
for (i=0; i<arr_size; i++)
if (compare(ptr + i*ele_size, x)) //L3
return i;
return -1;
}

int main()
{
int arr[] = {2, 5, 7, 90, 70};
int n = sizeof(arr)/sizeof(arr[0]);
int x = 7;
printf ("The index is %d ", search(arr, n, sizeof(int), &x, compare));
return 0;
}
```



Answer Options

Select any one option

Clear Answer

The index is 2

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Digital ends in 43 min

Aptitude

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Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Software

Starts in 43 min

Go to

Submit test

16. Working with arrays

‹ Previous Next ›

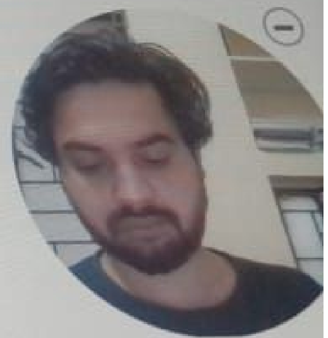
```
    return i;
  return -1;
}

int main()
{
 int arr[] = {2, 5, 7, 90, 70};
 int n = sizeof(arr)/sizeof(arr[0]);
 int x = 7;
 printf ("The index is %d ", search(arr, n, sizeof(int), &x, compare));
 return 0;
}
```

Answer Options

Select any one option

Clear Answer

- The index is 2
- The code will throw an error at L1 and L2
- The code will throw an error at L2
- The code will throw an error at L2 and L3



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Digital ends in 43 min

Aptitude

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Software

Starts in 43 min

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17. No. of cycles for high observability

< Previous | Next >

You notice that the observability of a certain system is high.

What does this indicate about the number of cycles that are required to measure the output node value?

Answer Options

Select any one option

Clear Answer

- It is more
- It is equal
- It is less
- None of these

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Digital ends in 43 min

Aptitude

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Software

Starts in 43 min

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18. Infinite loops

< Previous   Next >

```
module dummy;
logic [2:0] x, y, z, add, sub;
int a;
sum_difference dut (.*);
initial begin
fork
begin: add_dummy
for (a = 0; a < 20; a++)
begin
x = a;
y = a + 10;
#20 $display("This will add");
end
end
join_X

begin: sub_dummy
for (a = 10; a > 1; a--)
begin
z = a;
#10 $display("This will subtract);
end
end
endmodule
```

You find that the above given simulation runs into an infinite loop and does not produce the required output. What could have been the value of **X** in the given scenario?

Answer Options

Select any one option

Clear Answer

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Digital ends in 43 min

Aptitude

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Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Software

Starts in 43 min

Go to

Submit test

18. Infinite loops

Previous Next

```
begin: sub_dummy
for (a = 10; a > 1; a--)
begin
z = a;
#10 $display("This will subtract);
end
end
endmodule
```

*You find that the above given simulation runs into an infinite loop and does not produce the required output. What could have been the value of **X** in the given scenario?*

Answer Options

Select any one option

Clear Answer

- none
- any
- ✓ all Check
- Either choice 2 or 3

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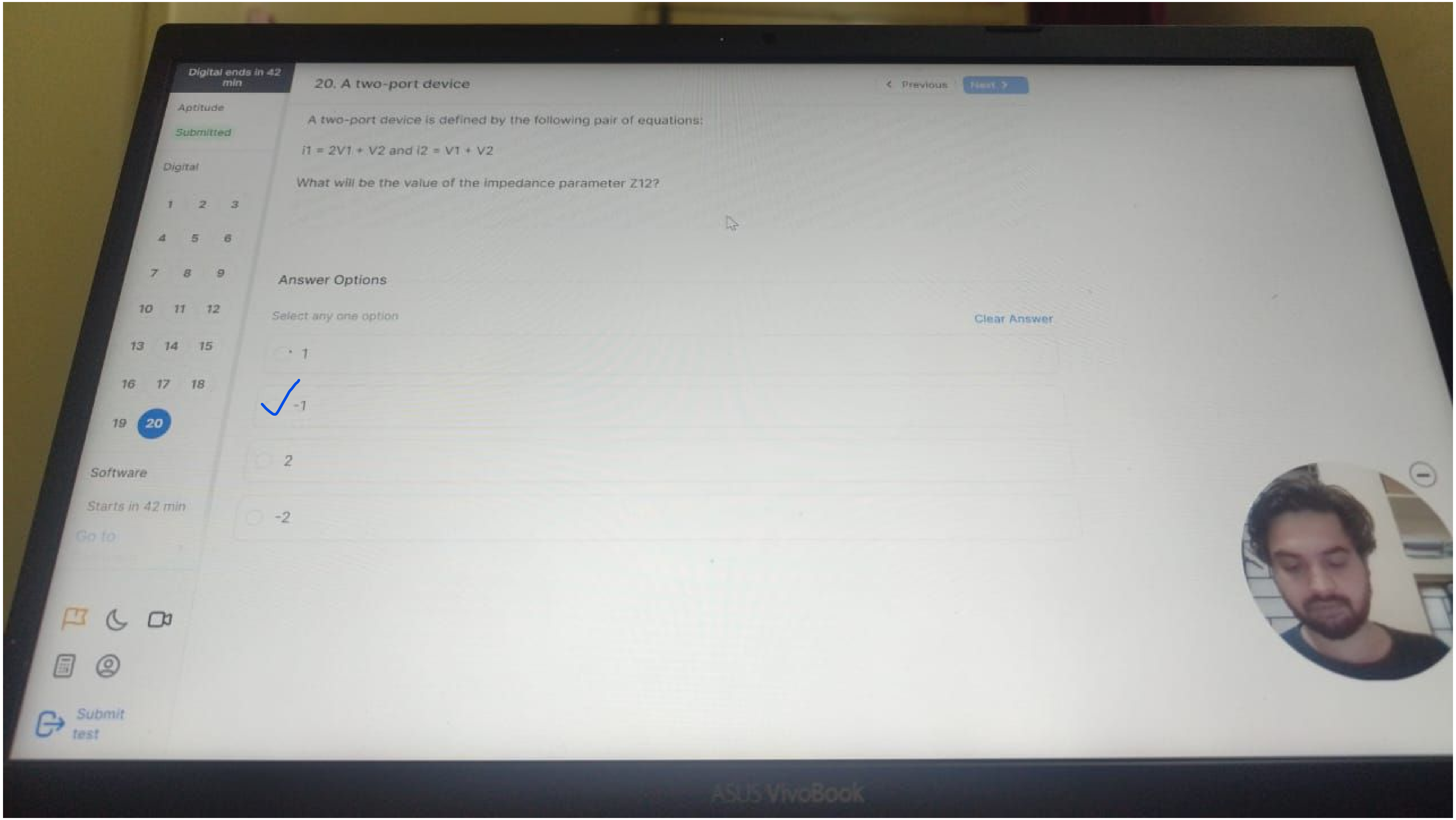
Digital ends in 42 min
Aptitude
Submitted
Digital
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Software
Starts in 42 min
Go to
Submit test
20. A two-port device
Previous
Next
A two-port device is defined by the following pair of equations:
i1 = 2V1 + V2 and i2 = V1 + V2
What will be the value of the impedance parameter Z12?
Answer Options
Select any one option
Clear Answer
1
-1
2
-2
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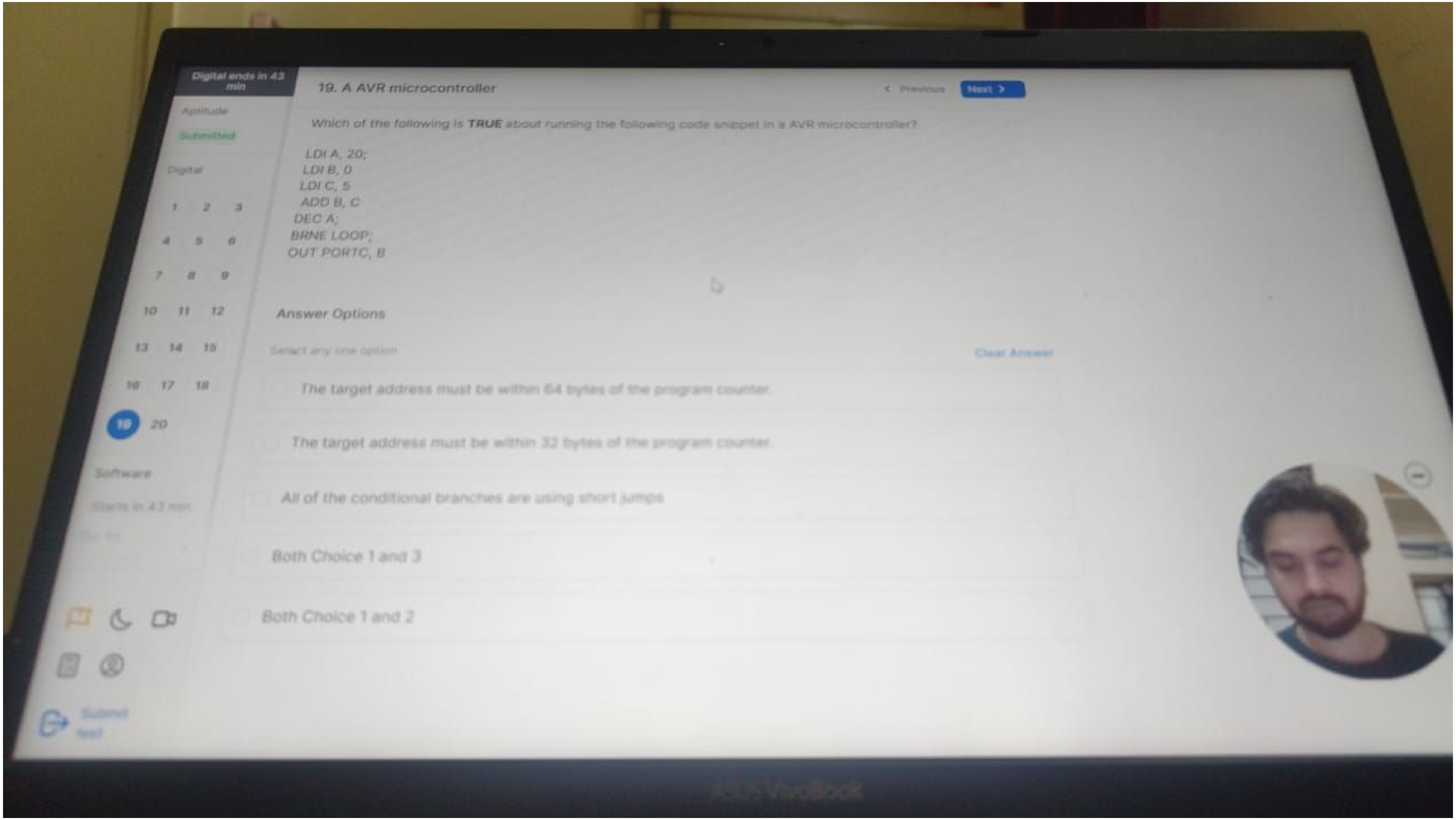
Digital ends in 43 min
Aptitude
Submitted
Digital
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Software
Starts in 43 min
Submit test
19. A AVR microcontroller
Previous
Next
Which of the following is TRUE about running the following code snippet in a AVR microcontroller?
LDI A, 20;
LDI B, 0
LDI C, 5
ADD B, C
DEC A;
BRNE LOOP;
OUT PORTC, B
Answer Options
Select any one option
Clear Answer
The target address must be within 64 bytes of the program counter.
The target address must be within 32 bytes of the program counter.
All of the conditional branches are using short jumps
Both Choice 1 and 3
Both Choice 1 and 2
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